Week 5: aquatic invertebrates

Individual assignment 5

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1. Set up

Packages

```
#Loading general use and cleaning packages
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

#Loading package for wrapping long axis labels
library(scales)

#Loading package for reading Excel files
library(readxl)

#Loading package for visualizing missing data
library(naniar)

#Loading package for arranging plots
library(patchwork)
```

Data

```r
#Reading in taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

#Reading in aquatic invertebrate data
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                    sheet = "Aquatic Insects")

#Reading in site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
                   sheet = "sites")

#Reading in water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
 ↪  Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Cleaning

```r
#Creating a new object from site information
ncos_sites <- sites |>
  #Filtering to only include NCOS sites sampled quarterly
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```r
#Creating a clean taxon list for joining with aquatic invertebrate data
taxon_list_clean <- taxon_list |>
  #Converting taxon names to snake case so they match the aquatic
   ↪  invertebrate columns
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))
```

```r
#Creating a clean water quality object
water_quality_clean <- water_quality |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only water quality surveys from quarterly NCOS sites
  semi_join(ncos_sites, by = "site") |>
  #Creating a date-site column for joining with aquatic invertebrate data
  unite("date_site", date, site, remove = TRUE) |>
  #Grouping by date-site survey
```

```r
  group_by(date_site) |>
  #Calculating median pH, salinity, and dissolved oxygen for each survey
  summarize(med_ph = median(p_h, na.rm = TRUE),
            med_sal = median(salinity_ppt, na.rm = TRUE),
            med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  #Ungrouping the dataframe
  ungroup() |>
  #Filtering out the salinity outlier identified in class
  filter(date_site != "2022-08-12_NVBR")
```

```r
#Creating a clean aquatic invertebrate object
aq_ins_clean <- aq_ins |>
  #Cleaning column names into snake case
  clean_names() |>
  #Keeping only sites that occur in the NCOS sites object
  semi_join(ncos_sites, by = c("site")) |>
  #Renaming the vial date column to date
  rename(date = date_on_vial) |>
  #Selecting survey information and focal aquatic invertebrate taxa
  select(date, sample_type, site,
         ostracod,
         copepod,
         hemiptera_corixidae_boatman,
         cladocera,
         nematode,
         diptera_ceratopogonidae,
         annelida_oligochaete,
         diptera_chironomid,
         annelida_polychaete) |>
  #Keeping only filtered beaker samples
  filter(sample_type %in% c("FB 250", "FB250")) |>
  #Converting taxon columns into long format
  pivot_longer(cols = ostracod:annelida_polychaete,
               names_to = "taxon",
               values_to = "abundance") |>
  #Grouping by date, site, and taxon
  group_by(date, site, taxon) |>
  #Calculating average abundance per liter from the 7.5 liter sample volume
  summarize(ave_lit = sum(abundance, na.rm = TRUE) / 7.5) |>
  #Ungrouping the dataframe
  ungroup() |>
  #Creating a date-site column for joining with water quality data
```

```
unite("date_site", date, site, remove = FALSE) |>
#Joining water quality information to each survey
left_join(water_quality_clean, by = c("date_site")) |>
#Joining taxonomic information to each taxon
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
#Adding full site names for clearer figure labels
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~  "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
#Ordering sites by median salinity so site patterns are easier to compare
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
 ↪  = TRUE),
       site_full = fct_rev(site_full))
```

Problems

1. Visualizing differences across sites in major

taxon abundance

a. Plan your figure

- X-Axis: `site_full`
- Y-Axis: log + 1 transformed average Ostracod abundance per liter
- Geometry to represent average abundance/L: `geom_jitter()`
- Geometry to represent medians: `stat_summary()`

b. Wrangle your data

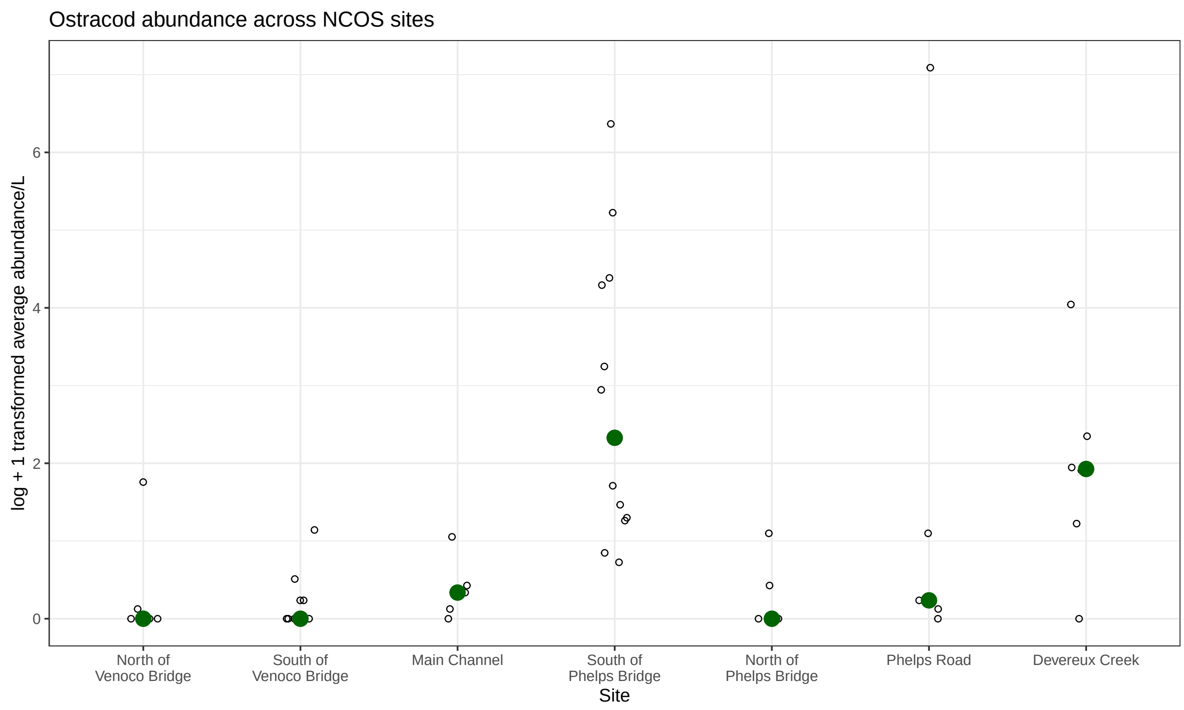
```r
#Creating a new object from clean aquatic invertebrate data
ostracods <- aq_ins_clean |>
  #Filtering to only include ostracods
  filter(taxon == "ostracod") |>
  #Log + 1 transforming average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

#Displaying 10 random rows from the ostracod data frame
ostracods |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
   date_site       date                site  taxon ave_lit med_ph med_sal med_do
   <chr>           <dttm>              <chr> <chr>   <dbl>  <dbl>   <dbl>  <dbl>
 1 2023-07-17_NPB  2023-07-17 00:00:00 NPB   ostr~   4.53   NA      NA      NA
 2 2022-11-12_NEC  2022-11-12 00:00:00 NEC   ostr~   0       7.33    7.01   2.4
 3 2023-05-04_NMC  2023-05-04 00:00:00 NMC   ostr~   1.87    8.32   NA      10.3
 4 2022-08-15_NMC  2022-08-15 00:00:00 NMC   ostr~   0.133   7.61   NA      NA
 5 2022-05-13_NVBR 2022-05-13 00:00:00 NVBR  ostr~   0       9.72   74.2    11.4
 6 2022-05-06_NMC  2022-05-06 00:00:00 NMC   ostr~   0       9.96    1.45   16.5
 7 2022-11-19_NDC  2022-11-19 00:00:00 NDC   ostr~   6      NA      NA      NA
 8 2020-02-21_NVBR 2020-02-21 00:00:00 NVBR  ostr~   4.8     7.34    4.1     6.1
 9 2021-11-12_NPB  2021-11-12 00:00:00 NPB   ostr~  79.2    NA      NA      NA
10 2023-03-03_NPB2 2023-03-03 00:00:00 NPB2  ostr~   0.133   7.71    1.98    9.59
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```r
#Creating an ostracod abundance figure across sites
ostracods_plot <- ggplot(data = ostracods,
                         #Setting x-axis to site and y-axis to
                          ↪  log-transformed abundance
                         mapping = aes(x = site_full,
                                       y = log_ave_lit)) +
  #Adding jittered points to show individual observations
  geom_jitter(height = 0,
              shape = 21,
```

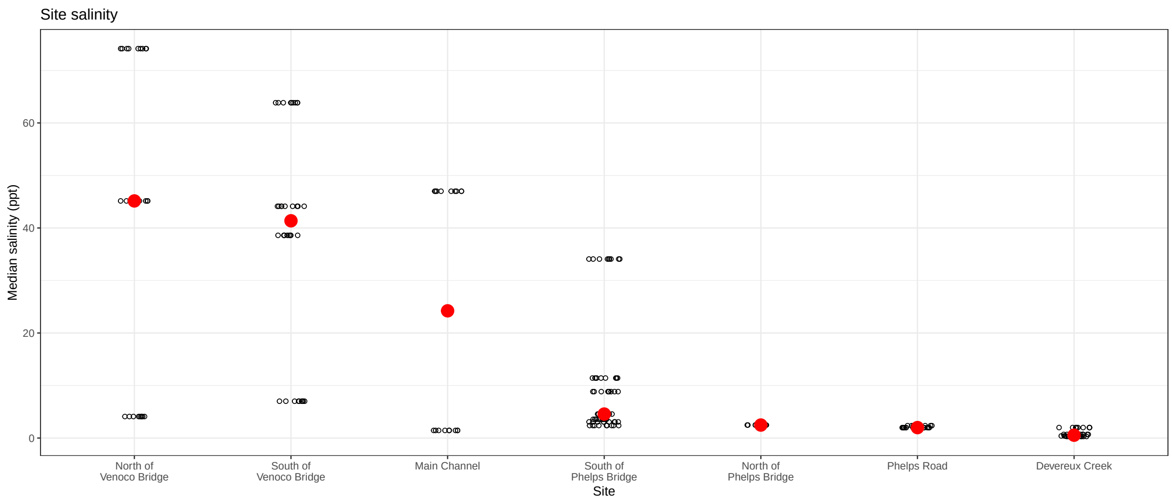
```
                 width = 0.1) +
  #Adding larger points to show median ostracod abundance at each site
  stat_summary(geom = "point",
               fun = "median",
               size = 4,
               color = "darkgreen") +
  #Wrapping long site names on the x-axis
  scale_x_discrete(labels = label_wrap(15)) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Ostracod abundance across NCOS sites") +
  #Using a different theme from default
  theme_bw()

#Displaying ostracod abundance figure
ostracods_plot
```

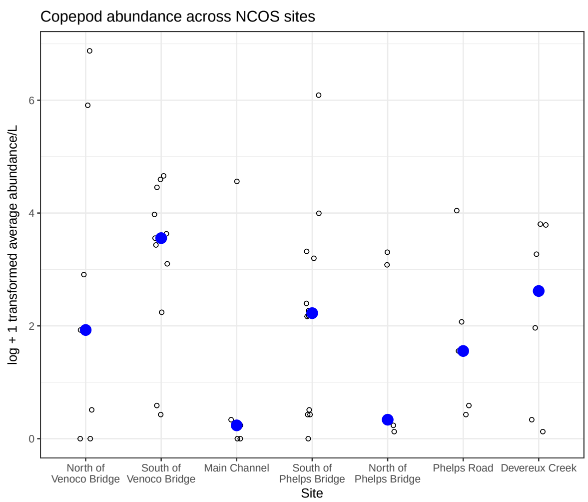


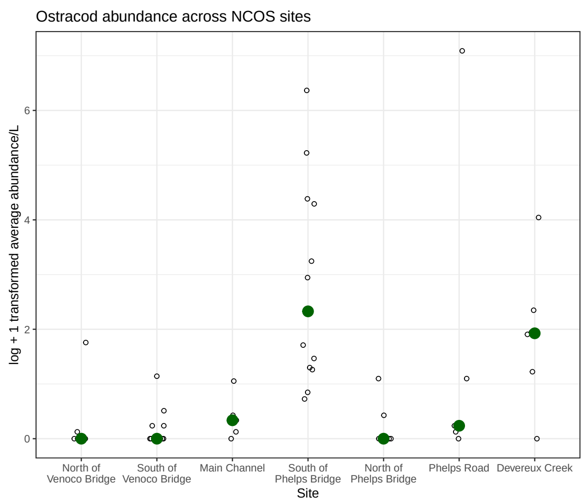
d. Create a multipanel figure

```
#Creating salinity figure from clean aquatic invertebrate data
salinity <- ggplot(data = aq_ins_clean,
                   #Setting x-axis to site and y-axis to median salinity
                   mapping = aes(x = site_full,
                                 y = med_sal)) +
  #Adding jittered points to show salinity values from each survey
  geom_jitter(height = 0,
              width = 0.1,
              shape = 21) +
  #Wrapping long site labels on the x-axis
  scale_x_discrete(labels = label_wrap(width = 15)) +
  #Adding red points to show median salinity at each site
  stat_summary(fun = median,
               color = "red",
               size = 1) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "Median salinity (ppt)",
       title = "Site salinity") +
  #Using a different theme from default
  theme_bw()

#Creating a clean copepod object from aquatic invertebrate data
copepods <- aq_ins_clean |>
  #Filtering to only include copepods
  filter(taxon == "copepod") |>
  #Log + 1 transforming average abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))
#Creating copepod abundance figure across sites
copepods_plot <- ggplot(data = copepods,
                        #Setting x-axis to site and y-axis to log-transformed
                        ↪  abundance
                        mapping = aes(x = site_full,
                                      y = log_ave_lit)) +
  #Adding jittered points to show individual observations
  geom_jitter(height = 0,
              shape = 21,
              width = 0.1) +
  #Adding blue points to show median copepod abundance at each site
  stat_summary(geom = "point",
```

```
               fun = "median",
               size = 4,
               color = "blue") +
  #Wrapping long site labels on the x-axis
  scale_x_discrete(labels = label_wrap(15)) +
  #Relabeling axes and adding a title
  labs(x = "Site",
       y = "log + 1 transformed average abundance/L",
       title = "Copepod abundance across NCOS sites") +
  #Using a different theme from default
  theme_bw()

#Combining salinity, copepod, and ostracod figures into one multipanel plot
salinity / (copepods_plot | ostracods_plot)
```


Site salinity
Median salinity (ppt)
0
20
40
60
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site


Copepod abundance across NCOS sites
log + 1 transformed average abundance/L
0
2
4
6
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site


Ostracod abundance across NCOS sites
log + 1 transformed average abundance/L
0
2
4
6
North of Venoco Bridge
South of Venoco Bridge
Main Channel
South of Phelps Bridge
North of Phelps Bridge
Phelps Road
Devereux Creek
Site